

Question ID 83f2c3bf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: 83f2c3bf

SAT3 M 3.1

$y = x + 4$

Which table gives three values of x and their corresponding values of y for the given equation?

A.

x	y
0	4
1	5
2	6

B.

x	y
0	6
1	5
2	4

C.

x	y
0	2
1	1
2	0

D.

x	y
0	0
1	1
2	2

ID: 83f2c3bf Answer

Correct Answer: A

Rationale

Choice A is correct. Substituting 0 for x into the given equation yields $y = 0 + 4$, or $y = 4$. Therefore, when $x = 0$, the corresponding value of y for the given equation is 4. Substituting 1 for x into the given equation yields $y = 1 + 4$, or $y = 5$. Therefore, when $x = 1$, the corresponding value of y for the given equation is 5. Substituting 2 for x into the given equation yields $y = 2 + 4$, or $y = 6$. Therefore, when $x = 2$, the

corresponding value of y for the given equation is **6**. Of the choices given, only the table in choice A gives these three values of x and their corresponding values of y for the given equation.

Choice B is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = -x + 6$.

Choice C is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = -x + 2$.

Choice D is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = x$.

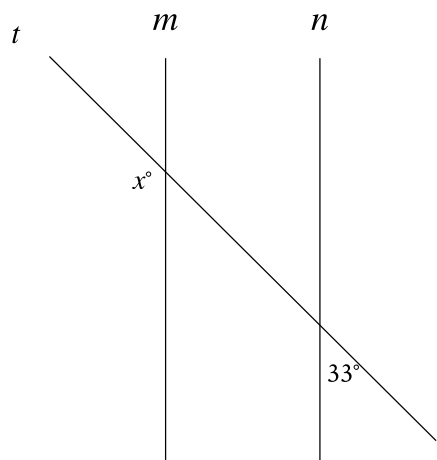
Question Difficulty: Easy

Question ID 0d3f51dc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 0d3f51dc

SAT3 M 3.2



Note: Figure not drawn to scale.

In the figure, line *m* is parallel to line *n*, and line *t* intersects both lines. What is the value of *x*?

- A. 33
- B. 57
- C. 123
- D. 147

ID: 0d3f51dc Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that line *m* is parallel to line *n*, and line *t* intersects both lines. It follows that line *t* is a transversal. When two lines are parallel and intersected by a transversal, exterior angles on the same side of the transversal are supplementary. Thus, $x + 33 = 180$. Subtracting 33 from both sides of this equation yields $x = 147$. Therefore, the value of *x* is 147.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID e9fb7774

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: e9fb7774

SAT3 M 3.3

What percentage of 300 is 75?

- A. 25%
- B. 50%
- C. 75%
- D. 225%

ID: e9fb7774 Answer

Correct Answer: A

Rationale

Choice A is correct. Let x represent the percentage of 300 that is 75. This can be written as $\frac{x}{100}(300) = 75$, or $3x = 75$. Dividing both sides of this equation by 3 yields $x = 25$. Therefore, 25% of 300 is 75.

Choice B is incorrect. 50% of 300 is 150, not 75.

Choice C is incorrect. 75% of 300 is 225, not 75.

Choice D is incorrect. 225% of 300 is 675, not 75.

Question Difficulty: Easy

Question ID 0b332f00

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 0b332f00

SAT3 M 3.4

The function g is defined by $g(x) = 6x$. For what value of x is $g(x) = 54$?

ID: 0b332f00 Answer

Correct Answer: 9

Rationale

The correct answer is 9. It’s given that $g(x) = 6x$. Substituting 54 for $g(x)$ in the given function yields $54 = 6x$. Dividing both sides of this equation by 6 yields $x = 9$. Therefore, the value of x when $g(x) = 54$ is 9.

Question Difficulty: Easy

Question ID 3c600337

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 3c600337

SAT3 M 3.5

The function f is defined by $f(x) = 270(0.1)^x$. What is the value of $f(0)$?

- A. 0
- B. 1
- C. 27
- D. 270

ID: 3c600337 Answer

Correct Answer: D

Rationale

Choice D is correct. The value of $f(0)$ is the value of $f(x)$ when $x = 0$. Substituting 0 for x in the given function yields $f(0) = 270(0.1)^0$, or $f(0) = 270(1)$, which is equivalent to $f(0) = 270$. Therefore, the value of $f(0)$ is 270.

Choice A is incorrect. This is the value of x , not $f(x)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $f(1)$, not $f(0)$.

Question Difficulty: Medium

Question ID 33e29881

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 33e29881

SAT3 M 3.6

In right triangle RST , the sum of the measures of angle R and angle S is 90 degrees. The value of $\sin(R)$ is $\frac{\sqrt{15}}{4}$. What is the value of $\cos(S)$?

- A. $\frac{\sqrt{15}}{15}$
- B. $\frac{\sqrt{15}}{4}$
- C. $\frac{4\sqrt{15}}{15}$
- D. $\sqrt{15}$

ID: 33e29881 Answer

Correct Answer: B

Rationale

Choice B is correct. The sine of any acute angle is equal to the cosine of its complement. It's given that in right triangle RST , the sum of the measures of angle R and angle S is 90 degrees. Therefore, angle R and angle S are complementary, and the value of $\sin R$ is equal to the value of $\cos S$. It's given that the value of $\sin R$ is $\frac{\sqrt{15}}{4}$, so the value of $\cos S$ is also $\frac{\sqrt{15}}{4}$.

Choice A is incorrect. This is the value of $\tan S$.

Choice C is incorrect. This is the value of $\frac{1}{\cos S}$.

Choice D is incorrect. This is the value of $\frac{1}{\tan S}$.

Question Difficulty: Medium

Question ID 8490cc45

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	■ ■ ■

ID: 8490cc45

SAT3 M 3.7

The function f is defined by $f(x) = (-8)(2)^x + 22$. What is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

- A. $(0, 14)$
- B. $(0, 2)$
- C. $(0, 22)$
- D. $(0, -8)$

ID: 8490cc45 Answer

Correct Answer: A

Rationale

Choice A is correct. The y -intercept of the graph of $y = f(x)$ in the xy -plane occurs at the point on the graph where $x = 0$. In other words, when $x = 0$, the corresponding value of $f(x)$ is the y -coordinate of the y -intercept. Substituting 0 for x in the given equation yields $f(0) = (-8)(2)^0 + 22$, which is equivalent to $f(0) = (-8)(1) + 22$, or $f(0) = 14$. Thus, when $x = 0$, the corresponding value of $f(x)$ is 14. Therefore, the y -intercept of the graph of $y = f(x)$ in the xy -plane is $(0, 14)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This could be the y -intercept for $f(x) = (-8)(2)^x$, not $f(x) = (-8)(2)^x + 22$.

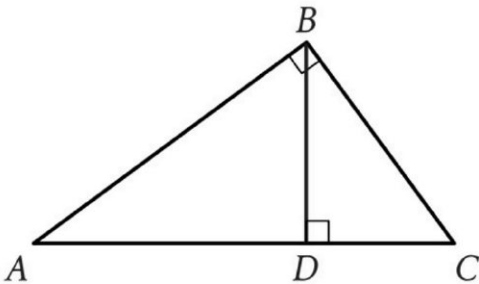
Question Difficulty: Hard

Question ID abcd0011

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	■ ■ ■

ID: abcd0011

SAT3 M 3.8



Note: Figure not drawn to scale.

In the figure shown, $AD = \frac{121}{3}$ and $AB = \frac{11\sqrt{130}}{3}$. What is the length of $\overline{DC}$?

Correct Answer: 3

Explanation

The correct answer is **3**. In the figure, $\angle ABC$ and $\angle BDC$ are right angles and are thus congruent. $\angle ACB$ and $\angle DCB$ are the same angle and are thus congruent. Therefore, $\triangle ABC$ is similar to $\triangle BDC$. Since $\angle BDC$ is a right angle, it follows that $\angle ADB$ is also a right angle. $\angle BAC$ and $\angle BAD$ are the same angle and are thus congruent. Therefore, $\triangle ABC$ is similar to $\triangle ADB$. Since $\triangle ABC$ is similar to $\triangle BDC$ and $\triangle ABC$ is similar to $\triangle ADB$, it follows that $\triangle BDC$ is similar to $\triangle ADB$. The ratios of corresponding sides of similar triangles are equal. Therefore, $\frac{DC}{DB} = \frac{DB}{AD}$. It's given that $AD = \frac{121}{3}$ and $AB = \frac{11\sqrt{130}}{3}$. By the Pythagorean theorem, the length of $\overline{DB}$ can be found using the equation $a^2 + b^2 = c^2$, where a and b are the lengths of the legs and c is the length of the hypotenuse of a right triangle. Substituting $\frac{121}{3}$ for a , DB for b , and $\frac{11\sqrt{130}}{3}$ for c in this equation yields $\left(\frac{121}{3}\right)^2 + (DB)^2 = \left(\frac{11\sqrt{130}}{3}\right)^2$, or $\frac{14,641}{9} + (DB)^2 = \frac{15,730}{9}$. Subtracting $\frac{14,641}{9}$ from both sides of this equation yields $(DB)^2 = \frac{1,089}{9}$. Taking the square root of both sides of this equation yields $DB = \pm\sqrt{\frac{1,089}{9}}$, or $DB = \pm 11$. Since the length of a side of a triangle must be positive, the length of $\overline{DB}$ is 11. Substituting 11 for DB and $\frac{121}{3}$ for AD in the equation $\frac{DC}{DB} = \frac{DB}{AD}$ yields $\frac{DC}{11} = \frac{11}{\left(\frac{121}{3}\right)}$. Multiplying both sides of this equation by 11 yields $\frac{(11)(DC)}{11} = \frac{(11)(11)}{\left(\frac{121}{3}\right)}$, which is equivalent to $DC = \frac{121}{\left(\frac{121}{3}\right)}$, or $DC = 3$. Thus, the length of $\overline{DC}$ is **3**.

Question ID d135f4bf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: d135f4bf

SAT3 M 3.9

The function f is defined by $f(x) = (x - 6)(x - 2)(x + 6)$. In the xy -plane, the graph of $y = g(x)$ is the result of translating the graph of $y = f(x)$ up 4 units. What is the value of $g(0)$?

ID: d135f4bf Answer

Correct Answer: 76

Rationale

The correct answer is **76**. It's given that the graph of $y = g(x)$ is the result of translating the graph of $y = f(x)$ up 4 units in the xy -plane. It follows that the graph of $y = g(x)$ is the same as the graph of $y = f(x) + 4$. Substituting $g(x)$ for y in the equation $y = f(x) + 4$ yields $g(x) = f(x) + 4$. It's given that $f(x) = (x - 6)(x - 2)(x + 6)$. Substituting $(x - 6)(x - 2)(x + 6)$ for $f(x)$ in the equation $g(x) = f(x) + 4$ yields $g(x) = (x - 6)(x - 2)(x + 6) + 4$. Substituting 0 for x in this equation yields $g(0) = (0 - 6)(0 - 2)(0 + 6) + 4$, or $g(0) = 76$. Thus, the value of $g(0)$ is **76**.

Question Difficulty: Hard

Question ID ba8ca563

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	■ ■ ■

ID: ba8ca563

SAT3 M 3.10

A cube has a volume of **474,552** cubic units. What is the surface area, in square units, of the cube?

ID: ba8ca563 Answer

Correct Answer: 36504

Rationale

The correct answer is **36,504**. The volume of a cube can be found using the formula $V = s^3$, where s represents the edge length of a cube. It's given that this cube has a volume of **474,552** cubic units. Substituting **474,552** for V in $V = s^3$ yields $474,552 = s^3$. Taking the cube root of both sides of this equation yields $78 = s$. Thus, the edge length of the cube is **78** units. Since each face of a cube is a square, it follows that each face has an edge length of **78** units. The area of a square can be found using the formula $A = s^2$. Substituting **78** for s in this formula yields $A = 78^2$, or $A = 6,084$. Therefore, the area of one face of this cube is **6,084** square units. Since a cube has **6** faces, the surface area, in square units, of this cube is **6(6,084)**, or **36,504**.

Question Difficulty: Hard

Question ID 1853bb35

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	■ ■ ■

ID: 1853bb35

SAT3 M 3.11

For the function q , the value of $q(x)$ decreases by 45% for every increase in the value of x by 1. If $q(0) = 14$, which equation defines q ?

- A. $q(x) = 0.55(14)^x$
- B. $q(x) = 1.45(14)^x$
- C. $q(x) = 14(0.55)^x$
- D. $q(x) = 14(1.45)^x$

ID: 1853bb35 Answer

Correct Answer: C

Rationale

Choice C is correct. Since the value of $q(x)$ decreases by a fixed percentage, 45%, for every increase in the value of x by 1, the function q is a decreasing exponential function. A decreasing exponential function can be written in the form $q(x) = a\left(1 - \frac{p}{100}\right)^x$, where a is the value of $q(0)$ and the value of $q(x)$ decreases by $p\%$ for every increase in the value of x by 1. If $q(0) = 14$, then $a = 14$. Since the value of $q(x)$ decreases by 45% for every increase in the value of x by 1, $p = 45$. Substituting 14 for a and 45 for p in the equation $q(x) = a\left(1 - \frac{p}{100}\right)^x$ yields $q(x) = 14\left(1 - \frac{45}{100}\right)^x$, which is equivalent to $q(x) = 14(1 - 0.45)^x$, or $q(x) = 14(0.55)^x$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. For this function, the value of $q(x)$ increases, rather than decreases, by 45% for every increase in the value of x by 1.

Question Difficulty: Hard

Question ID d8539e09

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	■ ■ ■

ID: d8539e09

SAT3 M 3.12

$y < 6x + 2$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
3	20
5	32
7	44

B.

x	y
3	16
5	36
7	40

C.

x	y
3	16
5	28
7	40

D.

x	y
3	24
5	36
7	48

ID: d8539e09 Answer

Correct Answer: C

Rationale

Choice C is correct. All the tables in the choices have the same three values of x , so each of the three values of x can be substituted in the given inequality to compare the corresponding values of y in each of the tables.

Substituting **3** for x in the given inequality yields $y < 6(3) + 2$, or $y < 20$. Therefore, when $x = 3$, the corresponding value of y is less than **20**. Substituting **5** for x in the given inequality yields $y < 6(5) + 2$, or $y < 32$. Therefore, when $x = 5$, the corresponding value of y is less than **32**. Substituting **7** for x in the given inequality yields $y < 6(7) + 2$, or $y < 44$. Therefore, when $x = 7$, the corresponding value of y is less than **44**. For the table in choice C, when $x = 3$, the corresponding value of y is **16**, which is less than **20**; when $x = 5$, the corresponding value of y is **28**, which is less than **32**; when $x = 7$, the corresponding value of y is **40**, which is less than **44**. Therefore, the table in choice C gives values of x and their corresponding values of y that are all solutions to the given inequality.

Choice A is incorrect. In the table for choice A, when $x = 3$, the corresponding value of y is **20**, which is not less than **20**; when $x = 5$, the corresponding value of y is **32**, which is not less than **32**; when $x = 7$, the corresponding value of y is **44**, which is not less than **44**.

Choice B is incorrect. In the table for choice B, when $x = 5$, the corresponding value of y is **36**, which is not less than **32**.

Choice D is incorrect. In the table for choice D, when $x = 3$, the corresponding value of y is **24**, which is not less than **20**; when $x = 5$, the corresponding value of y is **36**, which is not less than **32**; when $x = 7$, the corresponding value of y is **48**, which is not less than **44**.

Question Difficulty: Hard

Question ID c3b116d7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	■ ■ ■

ID: c3b116d7

SAT3 M 3.13

Which of the following expressions is(are) a factor of $3x^2 + 20x - 63$?

I. $x - 9$

II. $3x - 7$

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

ID: c3b116d7 Answer

Correct Answer: B

Rationale

Choice B is correct. The given expression can be factored by first finding two values whose sum is **20** and whose product is **$3(-63)$** , or **-189** . Those two values are **27** and **-7** . It follows that the given expression can be rewritten as **$3x^2 + 27x - 7x - 63$** . Since the first two terms of this expression have a common factor of **$3x$** and the last two terms of this expression have a common factor of **-7** , this expression can be rewritten as **$3x(x + 9) - 7(x + 9)$** . Since the two terms of this expression have a common factor of **$(x + 9)$** , it can be rewritten as **$(3x - 7)(x + 9)$** . Therefore, expression II, **$3x - 7$** , is a factor of **$3x^2 + 20x - 63$** , but expression I, **$x - 9$** , is not a factor of **$3x^2 + 20x - 63$** .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 14360f84

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	■ ■ ■

ID: 14360f84

SAT3 M 3.14

$$\begin{aligned}4x - 9y &= 9y + 5 \\ hy &= 2 + 4x\end{aligned}$$

In the given system of equations, h is a constant. If the system has no solution, what is the value of h ?

- A. -9
- B. 0
- C. 9
- D. 18

ID: 14360f84 Answer

Correct Answer: D

Rationale

Choice D is correct. A system of two linear equations in two variables, x and y , has no solution if the lines represented by the equations in the xy -plane are distinct and parallel. The graphs of two lines in the xy -plane represented by equations in the form $Ax + By = C$, where A , B , and C are constants, are parallel if the coefficients for x and y in one equation are proportional to the corresponding coefficients in the other equation. The first equation in the given system can be written in the form $Ax + By = C$ by subtracting $9y$ from both sides of the equation to yield $4x - 18y = 5$. The second equation in the given system can be written in the form $Ax + By = C$ by subtracting $4x$ from both sides of the equation to yield $-4x + hy = 2$. The coefficient of x in this second equation, -4 , is -1 times the coefficient of x in the first equation, 4 . For the lines to be parallel, the coefficient of y in the second equation, h , must also be -1 times the coefficient of y in the first equation, -18 . Thus, $h = -1(-18)$, or $h = 18$. Therefore, if the given system has no solution, the value of h is 18 .

Choice A is incorrect. If the value of h is -9 , then the given system would have one solution, rather than no solution.

Choice B is incorrect. If the value of h is 0 , then the given system would have one solution, rather than no solution.

Choice C is incorrect. If the value of h is 9 , then the given system would have one solution, rather than no solution.

Question Difficulty: Hard

Question ID abcd0012

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	■ ■ ■

ID: abcd0012

SAT3 M 3.15

$$(3x + p)(5x^2 - 45)(2x^2 - 16x + 6p) = 0$$

In the given equation, p is a positive constant. The sum of the solutions to the equation is $\frac{20}{3}$. What is the value of p ?

Correct Answer: 4

Explanation

The correct answer is 4. For the given equation, p is a positive constant and the sum of the solutions to the equation is $\frac{20}{3}$. Applying the zero product property to the given equation yields three equations: $3x + p = 0$, $5x^2 - 45 = 0$, and $2x^2 - 16x + 6p = 0$. Subtracting p from both sides of the equation $3x + p = 0$ yields $3x = -p$. Dividing both sides of this equation by 3 yields $x = -\frac{p}{3}$. Adding 45 to both sides of the equation $5x^2 - 45 = 0$ yields $5x^2 = 45$. Dividing both sides of this equation by 5 yields $x^2 = 9$. Taking the square root of both sides of this equation yields two solutions: $x = -3$ and $x = 3$. Dividing both sides of the equation $2x^2 - 16x + 6p = 0$ by 2 yields $x^2 - 8x + 3p = 0$. A quadratic equation in the form $ax^2 + bx + c = 0$ can be solved using the formula $x = \frac{-b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a}$. For the equation $x^2 - 8x + 3p = 0$, $a = 1$, $b = -8$, and $c = 3p$. Therefore, the solutions to $x^2 - 8x + 3p = 0$ are $x = \frac{8}{2(1)} + \frac{\sqrt{(-8)^2 - 4(1)(3p)}}{2(1)}$ and $x = \frac{8}{2(1)} - \frac{\sqrt{(-8)^2 - 4(1)(3p)}}{2(1)}$, which can be rewritten as $x = 4 + \sqrt{16 - 3p}$ and $x = 4 - \sqrt{16 - 3p}$, respectively. Thus, the sum of the solutions to the equation $(3x + p)(5x^2 - 45)(2x^2 - 16x + 6p) = 0$ can be represented by $-\frac{p}{3} + 3 - 3 + 4 + \sqrt{16 - 3p} + 4 - \sqrt{16 - 3p}$, or $-\frac{p}{3} + 8$. Since it's given that the sum of the solutions is $\frac{20}{3}$, it follows that $-\frac{p}{3} + 8 = \frac{20}{3}$. Subtracting 8 from both sides of this equation yields $-\frac{p}{3} = -\frac{4}{3}$. It follows that the value of p is 4.

Question ID ee7b1de1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	■ ■ ■

ID: ee7b1de1

SAT3 M 3.16

A small business owner budgets **\$2,200** to purchase candles. The owner must purchase a minimum of **200** candles to maintain the discounted pricing. If the owner pays **\$4.90** per candle to purchase small candles and **\$11.60** per candle to purchase large candles, what is the maximum number of large candles the owner can purchase to stay within the budget and maintain the discounted pricing?

ID: ee7b1de1 Answer

Correct Answer: 182

Rationale

The correct answer is **182**. Let s represent the number of small candles the owner can purchase, and let ℓ represent the number of large candles the owner can purchase. It's given that the owner pays **\$4.90** per candle to purchase small candles and **\$11.60** per candle to purchase large candles. Therefore, the owner pays **4.90s** dollars for s small candles and **11.60ℓ** dollars for ℓ large candles, which means the owner pays a total of **4.90s + 11.60ℓ** dollars to purchase candles. It's given that the owner budgets **\$2,200** to purchase candles. Therefore, **4.90s + 11.60ℓ ≤ 2,200**. It's also given that the owner must purchase a minimum of **200** candles. Therefore, **s + ℓ ≥ 200**. The inequalities **4.90s + 11.60ℓ ≤ 2,200** and **s + ℓ ≥ 200** can be combined into one compound inequality by rewriting the second inequality so that its left-hand side is equivalent to the left-hand side of the first inequality. Subtracting ℓ from both sides of the inequality **s + ℓ ≥ 200** yields **s ≥ 200 - ℓ**. Multiplying both sides of this inequality by **4.90** yields **4.90s ≥ 4.90(200 - ℓ)**, or **4.90s ≥ 980 - 4.90ℓ**. Adding **11.60ℓ** to both sides of this inequality yields **4.90s + 11.60ℓ ≥ 980 - 4.90ℓ + 11.60ℓ**, or **4.90s + 11.60ℓ ≥ 980 + 6.70ℓ**. This inequality can be combined with the inequality **4.90s + 11.60ℓ ≤ 2,200**, which yields the compound inequality **980 + 6.70ℓ ≤ 4.90s + 11.60ℓ ≤ 2,200**. It follows that **980 + 6.70ℓ ≤ 2,200**. Subtracting **980** from both sides of this inequality yields **6.70ℓ ≤ 2,200**. Dividing both sides of this inequality by **6.70** yields approximately **ℓ ≤ 182.09**. Since the number of large candles the owner purchases must be a whole number, the maximum number of large candles the owner can purchase is the largest whole number less than **182.09**, which is **182**.

Question Difficulty: Hard

Question ID b7c74b73

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: b7c74b73

SAT3 M 3.17

$$f(x) = 5,470(0.64)^{\frac{x}{12}}$$

The function f gives the value, in dollars, of a certain piece of equipment after x months of use. If the value of the equipment decreases each year by $p\%$ of its value the preceding year, what is the value of p ?

- A. 4
- B. 5
- C. 36
- D. 64

ID: b7c74b73 Answer

Correct Answer: C

Rationale

Choice C is correct. For a function of the form $f(x) = a(r)^{\frac{x}{k}}$, where a , r , and k are constants and $r < 1$, the value of $f(x)$ decreases by $100(1 - r)\%$ for every increase of x by k . In the given function, $a = 5,470$, $r = 0.64$, and $k = 12$. Therefore, for the given function, the value of $f(x)$ decreases by $100(1 - 0.64)\%$, or **36%**, for every increase of x by **12**. Since $f(x)$ represents the value, in dollars, of the equipment after x months of use, it follows that the value of the equipment decreases every **12 months** by **36%** of its value the preceding **12 months**. Since there are **12 months** in a year, the value of the equipment decreases each year by **36%** of its value the preceding year. Thus, the value of p is **36**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 0e61101e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	■ ■ ■

ID: 0e61101e

SAT3 M 3.18

$$f(x) = 9(4)^x$$

The function f is defined by the given equation. If $g(x) = f(x + 2)$, which of the following equations defines the function g ?

- A. $g(x) = 18(4)^x$
- B. $g(x) = 144(4)^x$
- C. $g(x) = 18(8)^x$
- D. $g(x) = 81(16)^x$

ID: 0e61101e Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that $f(x) = 9(4)^x$ and $g(x) = f(x + 2)$. Substituting $x + 2$ for x in $f(x) = 9(4)^x$ gives $f(x + 2) = 9(4)^{x+2}$. Rewriting this equation using properties of exponents gives $f(x + 2) = 9(4)^x(4)^2$, which is equivalent to $f(x + 2) = 9(4)^x(16)$. Multiplying 9 and 16 in this equation gives $f(x + 2) = 144(4)^x$. Since $g(x) = f(x + 2)$, $g(x) = 144(4)^x$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID ba0edc30

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	■ ■ ■

ID: ba0edc30

SAT3 M 3.19

$$x^2 - 2x - 9 = 0$$

One solution to the given equation can be written as $1 + \sqrt{k}$, where k is a constant. What is the value of k ?

- A. 8
- B. 10
- C. 20
- D. 40

ID: ba0edc30 Answer

Correct Answer: B

Rationale

Choice B is correct. Adding 9 to each side of the given equation yields $x^2 - 2x = 9$. To complete the square, adding 1 to each side of this equation yields $x^2 - 2x + 1 = 9 + 1$, or $(x - 1)^2 = 10$. Taking the square root of each side of this equation yields $x - 1 = \pm\sqrt{10}$. Adding 1 to each side of this equation yields $x = 1 \pm \sqrt{10}$. Since it's given that one of the solutions to the equation can be written as $1 + \sqrt{k}$, the value of k must be 10.

Alternate approach: It's given that $1 + \sqrt{k}$ is a solution to the given equation. It follows that $x = 1 + \sqrt{k}$.

Substituting $1 + \sqrt{k}$ for x in the given equation yields $(1 + \sqrt{k})^2 - 2(1 + \sqrt{k}) - 9 = 0$, or $(1 + \sqrt{k})(1 + \sqrt{k}) - 2(1 + \sqrt{k}) - 9 = 0$. Expanding the products on the left-hand side of this equation yields $1 + 2\sqrt{k} + k - 2 - 2\sqrt{k} - 9 = 0$, or $k - 10 = 0$. Adding 10 to each side of this equation yields $k = 10$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 03ff48d2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	■ ■ ■

ID: 03ff48d2

SAT3 M 3.20

$$x(kx - 56) = -16$$

In the given equation, k is an integer constant. If the equation has no real solution, what is the least possible value of k ?

ID: 03ff48d2 Answer

Correct Answer: 50

Rationale

The correct answer is **50**. An equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, has no real solutions if and only if its discriminant, $b^2 - 4ac$, is negative. Applying the distributive property to the left-hand side of the equation $x(kx - 56) = -16$ yields $kx^2 - 56x = -16$. Adding **16** to each side of this equation yields $kx^2 - 56x + 16 = 0$. Substituting k for a , -56 for b , and **16** for c in $b^2 - 4ac$ yields a discriminant of $(-56)^2 - 4(k)(16)$, or **3,136** $- 64k$. If the given equation has no real solution, it follows that the value of **3,136** $- 64k$ must be negative. Therefore, **3,136** $- 64k < 0$. Adding **64k** to both sides of this inequality yields **3,136** $< 64k$. Dividing both sides of this inequality by **64** yields **49** $< k$, or $k > 49$. Since it's given that k is an integer, the least possible value of k is **50**.

Question Difficulty: Hard

Question ID 8c5e6702

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	■ ■ ■

ID: 8c5e6702

SAT3 M 3.21

A window repair specialist charges **\$220** for the first two hours of repair plus an hourly fee for each additional hour. The total cost for **5** hours of repair is **\$400**. Which function f gives the total cost, in dollars, for x hours of repair, where $x \geq 2$?

- A. $f(x) = 60x + 100$
- B. $f(x) = 60x + 220$
- C. $f(x) = 80x$
- D. $f(x) = 80x + 220$

ID: 8c5e6702 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the window repair specialist charges **\$220** for the first two hours of repair plus an hourly fee for each additional hour. Let n represent the hourly fee for each additional hour after the first two hours. Since it's given that x is the number of hours of repair, it follows that the charge generated by the hourly fee after the first two hours can be represented by the expression $n(x - 2)$. Therefore, the total cost, in dollars, for x hours of repair is $f(x) = 220 + n(x - 2)$. It's given that the total cost for **5** hours of repair is **\$400**. Substituting **5** for x and **400** for $f(x)$ into the equation $f(x) = 220 + n(x - 2)$ yields $400 = 220 + n(5 - 2)$, or $400 = 220 + 3n$. Subtracting **220** from both sides of this equation yields $180 = 3n$. Dividing both sides of this equation by **3** yields $n = 60$. Substituting **60** for n in the equation $f(x) = 220 + n(x - 2)$ yields $f(x) = 220 + 60(x - 2)$, which is equivalent to $f(x) = 220 + 60x - 120$, or $f(x) = 60x + 100$. Therefore, the total cost, in dollars, for x hours of repair is $f(x) = 60x + 100$.

Choice B is incorrect. This function represents the total cost, in dollars, for x hours of repair where the specialist charges **\$340**, rather than **\$220**, for the first two hours of repair.

Choice C is incorrect. This function represents the total cost, in dollars, for x hours of repair where the specialist charges **\$160**, rather than **\$220**, for the first two hours of repair, and an hourly fee of **\$80**, rather than **\$60**, after the first two hours.

Choice D is incorrect. This function represents the total cost, in dollars, for x hours of repair where the specialist charges **\$380**, rather than **\$220**, for the first two hours of repair, and an hourly fee of **\$80**, rather than **\$60**, after the first two hours.

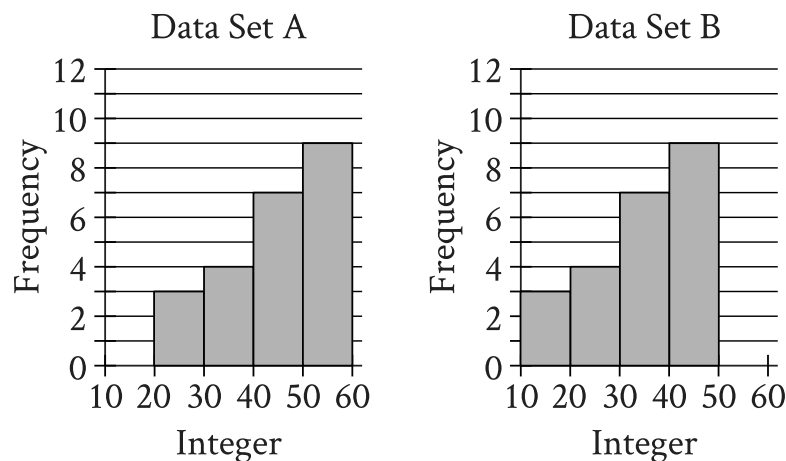
Question Difficulty: Hard

Question ID f8a322d9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	■ ■ ■

ID: f8a322d9

SAT3 M 3.22



Two data sets of **23** integers each are summarized in the histograms shown. For each of the histograms, the first interval represents the frequency of integers greater than or equal to **10**, but less than **20**. The second interval represents the frequency of integers greater than or equal to **20**, but less than **30**, and so on. What is the smallest possible difference between the mean of data set A and the mean of data set B?

- A. 0
- B. 1
- C. 10
- D. 23

ID: f8a322d9 Answer

Correct Answer: B

Rationale

Choice B is correct. The histograms shown have the same shape, but data set A contains values between **20** and **60** and data set B contains values between **10** and **50**. Thus, the mean of data set A is greater than the mean of data set B. Therefore, the smallest possible difference between the mean of data set A and the mean of data set B is the difference between the smallest possible mean of data set A and the greatest possible mean of data set B. In data set A, since there are **3** integers in the interval greater than or equal to **20** but less than **30**, **4** integers greater than or equal to **30** but less than **40**, **7** integers greater than or equal to **40** but less than **50**, and **9** integers greater than or equal to **50** but less than **60**, the smallest possible mean for data set A is $\frac{(3 \cdot 20) + (4 \cdot 30) + (7 \cdot 40) + (9 \cdot 50)}{23}$. In data set B, since there are **3** integers greater than or equal to **10** but less than **20**, **4** integers greater than or equal to **20** but less than **30**, **7** integers greater than or equal to **30** but less than **40**, and **9** integers greater than or equal to **40** but less than **50**, the largest possible mean for data set B is

$\frac{(3 \cdot 19) + (4 \cdot 29) + (7 \cdot 39) + (9 \cdot 49)}{23}$. Therefore, the smallest possible difference between the mean of data set A and the mean of data set B is $\frac{(3 \cdot 20) + (4 \cdot 30) + (7 \cdot 40) + (9 \cdot 50)}{23} - \frac{(3 \cdot 19) + (4 \cdot 29) + (7 \cdot 39) + (9 \cdot 49)}{23}$, which is equivalent to $\frac{(3 \cdot 20) - (3 \cdot 19) + (4 \cdot 30) - (4 \cdot 29) + (7 \cdot 40) - (7 \cdot 39) + (9 \cdot 50) - (9 \cdot 49)}{23}$. This expression can be rewritten as $\frac{3(20-19) + 4(30-29) + 7(40-39) + 9(50-49)}{23}$, or $\frac{23}{23}$, which is equal to 1. Therefore, the smallest possible difference between the mean of data set A and the mean of data set B is 1.

Choice A is incorrect. This is the smallest possible difference between the ranges, not the means, of the data sets.

Choice C is incorrect. This is the difference between the greatest possible mean, not the smallest possible mean, of data set A and the greatest possible mean of data set B.

Choice D is incorrect. This is the smallest possible difference between the sum of the values in data set A and the sum of the values in data set B, not the smallest possible difference between the means.

Question Difficulty: Hard